Tsai-Shien Chen

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Education

University of California, Merced — *Merced, CA*

Ph.D. candidate in Electrical Engineering and Computer Science — *Sep. 2022 - Now*

- Advisor: Prof. Mukesh Singhal

National Taiwan University — *Taipei, Taiwan*

M.S. in Electronics Engineering — *Sep. 2019 - March 2022*

- Advisor: Prof. Shao-Yi Chien

B.S. in Electrical Engineering — *Sep. 2015 - June 2019*

- GPA: 4.23 / 4.30; Valedictorian

Research Interests

- Image, Video, World Generative Models
- Video Understanding, Generation, and Application
- Controllable and Personalized Content Creation

Honors & Awards

2026	**NeurIPS Top Reviewer**
2026	**CVPR Outstanding Reviewer**
2026	**CVPR Doctoral Consortium**
2025	**Google PhD Fellowship**
2024	**Graduate Student Opportunity Program Fellowship (one year of full tuition and stipend)**, UC Merced
2023	**Best Master Thesis of the Year Award**, Graduate Institute of Electronics Engineering, NTU
2022	**Honorary Member (top 2% in college)**, Phi Tau Phi Scholastic Honor Society
2019	**Valedictorian**, Department of Electrical Engineering, NTU
2015-2019	**4-time Presidential Award (top 5% in department)**, NTU

Work Experiences

Adobe Research — *San Francisco, CA*

Research Intern — *March 2026 - Now*

- Mentors: Richard Zhang, Eli Shechtman
- Real-time autoregressive video streaming model

Snap Inc. — *Santa Monica, CA*

Research Intern — *May 2023 - March 2026*

- Mentors: Aliaksandr Siarohin, Sergey Tulyakov, Jun-Yan Zhu, Kfir Aberman
- Large-scale video dataset and video captioning, generation, and personalization models

Publications

Conference Proceedings

1. **CineOrchestra: Unified Entity-Centric Conditioning for Cinematic Video Generation**
   Sharath Girish*, **Tsai-Shien Chen***, Zhikang Dong, Mukesh Singhal, Hao Chen, Sergey Tulyakov, Aliaksandr Siarohin
   NeurIPS 2026
2. **CanvasComposer: Personalized Group Photo Generation via a Multi-Reference Canvas**
   Gordon Guocheng Qian, Ruihang Zhang, **Tsai-Shien Chen**, Yusuf Dalva, Anujraaj Argo Goyal, Willi Menapace, Ivan Skorokhodov, Meng Dong, Arpit Sahni, Daniil Ostashev, Ju Hu, Mukesh Singhal, Sergey Tulyakov, Kuan-Chieh Jackson Wang
   SIGGRAPH Asia 2026

3. **Canvas-to-Image: Compositional Image Generation with Multimodal Controls**
   Yusuf Dalva, Gordon Guocheng Qian, Maya Goldenberg, **Tsai-Shien Chen**, Kfir Aberman, Sergey Tulyakov, Pinar Yanardag, Kuan-Chieh Jackson Wang
   SIGGRAPH 2026
4. **Omni-Attribute: Open-vocabulary Attribute Encoder for Visual Concept Personalization**
   Tsai-Shien Chen, Aliaksandr Siarohin, Gordon Guocheng Qian, Kuan-Chieh Jackson Wang, Egor Nemchinov, Moayed Haji-Ali, Riza Alp Guler, Willi Menapace, Ivan Skorokhodov, Anil Kag, Jun-Yan Zhu, Sergey Tulyakov
   CVPR 2026
5. **AlcheMinT: Fine-grained Temporal Control for Multi-Reference Consistent Video Generation**
   Sharath Girish, Viacheslav Ivanov, **Tsai-Shien Chen**, Hao Chen, Aliaksandr Siarohin, Sergey Tulyakov
   CVPR 2026
6. **Multi-subject Open-set Personalization in Video Generation**
   Tsai-Shien Chen, Aliaksandr Siarohin, Willi Menapace, Yuwei Fang, Kwot Sin Lee, Ivan Skorokhodov, Kfir Aberman, Jun-Yan Zhu, Ming-Hsuan Yang, Sergey Tulyakov
   CVPR 2025
7. **VIMI: Grounding Video Generation through Multi-modal Instruction**
   Yuwei Fang, Willi Menapace, Aliaksandr Siarohin, **Tsai-Shien Chen**, Kuan-Chieh Jackson Wang, Ivan Skorokhodov, Graham Neubig, Sergey Tulyakov
   EMNLP 2024
8. **Panda-70M: Captioning 70M Videos with Multiple Cross-Modality Teachers**
   Tsai-Shien Chen, Aliaksandr Siarohin, Willi Menapace, Ekaterina Deyneka, Hsiang-wei Chao, Byung Eun Jeon, Yuwei Fang, Hsin-Ying Lee, Jian Ren, Ming-Hsuan Yang, Sergey Tulyakov
   CVPR 2024
9. **Snap Video: Scaled Spatiotemporal Transformers for Text-to-Video Synthesis**
   Willi Menapace, Aliaksandr Siarohin, Ivan Skorokhodov, Ekaterina Deyneka, **Tsai-Shien Chen**, Anil Kag, Yuwei Fang, Aleksei Stoliar, Elisa Ricci, Jian Ren, Sergey Tulyakov
   CVPR 2024 (Highlight Presentation)
10. **Incremental False Negative Detection for Contrastive Learning**
    Tsai-Shien Chen, Wei-Chih Hung, Hung-Yu Tseng, Shao-Yi Chien, Ming-Hsuan Yang
    ICLR 2022
11. **Hard Samples Rectification for Unsupervised Cross-domain Person Re-identification**
    Chih-Ting Liu, Man-Yu Lee, **Tsai-Shien Chen**, Shao-Yi Chien
    ICIP 2021
12. **Tracklet-Refined Multi-Camera Tracking Based on Balanced Cross-Domain Re-Identification for Vehicles**
    Kai-Siang Yang, Yu-Kai Chen, **Tsai-Shien Chen**, Chih-Ting Liu, Shao-Yi Chien
    CVPR Workshops 2021
13. **Orientation-aware Vehicle Re-identification with Semantics-guided Part Attention Network**
    Tsai-Shien Chen, Chih-Ting Liu, Chih-Wei Wu, Shao-Yi Chien
    ECCV 2020 (Oral Presentation)
14. **Viewpoint-Aware Channel-Wise Attentive Network for Vehicle Re-Identification**
    Tsai-Shien Chen, Man-Yu Lee, Chih-Ting Liu, Shao-Yi Chien
    CVPR Workshops 2020
15. **Supervised Joint Domain Learning for Vehicle Re-Identification**
    Chih-Ting Liu, Man-Yu Lee, Chih-Wei Wu, Bo-Ying Chen, **Tsai-Shien Chen**, Yao-Ting Hsu, Shao-Yi Chien
    CVPR Workshops 2019

Patents

1. **Captioning Videos with Multiple Cross-Modality Teachers**
   Tsai-Shien Chen, Yuwei Fang, Hsin-Ying Lee, Jian Ren, Aliaksandr Siarohin, Sergey Tulyakov
   US12489956B2
2. **Scaled Spatiotemporal Transformers for Text-to-Video Synthesis**
   Tsai-Shien Chen, Yuwei Fang, Anil Kag, Willi Menapace, Jian Ren, Aliaksandr Siarohin, Ivan Skorokhodov, Sergey Tulyakov
   US12700160B2

Professional Activities

Invited Talks	- **UC Merced** (Sep 2026): "Toward Controllable Video Generation: From Web-Scale Data to Efficient Cinematic Control"
	- **ByteDance** (April 2026): "From Data to Director: Towards Controllable and Personalized Movie Generation"
	- **AI Time** (Dec 2025): "From Personalization Models towards Movie Generator"
	- **Anuttacon** (July 2025): "Learning Video Generation and Personalization from Large-Scale Datasets"
Conf. Reviewer	- **CVPR**'22,'23,'24,'25,'26; **ICCV**'21,'23,'25; **ECCV**'22,'24,'26; **WACV**'26; **ACCV**'24
	- **NeurIPS**'23,'24,'25,'26; **ICLR**'24,'25,'26,'27; **ICML**'24,'25,'26; **AAAI**'26
	- **SIGGRAPH**'26; **SIGGRAPH Asia**'26; **ACMMM**'25,'26
Journal Reviewer	- IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**)
	- IEEE Transactions on Image Processing (**TIP**)
	- International Journal of Computer Vision (**IJCV**)
Teaching Assistant	- UC Merced: CSE-185 **Introduction to Computer Vision** (Spring 2023)
	- UC Merced: CSE-024 **Advanced Programming** (Fall 2022)
	- NTU: EEE-5053 **Computer Vision** (Spring 2021)